Assignment 7

AI Workflow Automation using LangGraph and Gemini

Overview

This assignment focuses on designing and implementing intelligent workflow automation systems using **LangGraph**, **LangChain**, and **Google Gemini 2.5 Flash**. The objective is to build structured multi-step decision-making workflows where rule-based validation and LLM-driven reasoning work together to automate real-world business processes.

Two practical enterprise use cases were implemented:

1. **Telecom SIM Activation Workflow**
2. **Healthcare Appointment Prioritization Workflow**

Both workflows use:

- Tool-based deterministic validation
- Gemini-powered reasoning and classification
- State-driven execution using LangGraph
- Final automated decision generation

Technology Stack

Component	Technology Used
Workflow Orchestration	LangGraph
LLM Framework	LangChain
Language Model	Google Gemini 2.5 Flash
Programming Language	Python
Tool Integration	LangChain Tools
Environment	Jupyter Notebook

Use Case 1: Telecom SIM Activation Workflow

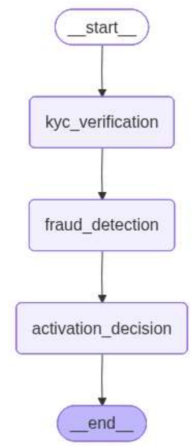
Problem Statement

Telecom providers must verify customer identity and detect fraudulent SIM requests before activation. Manual verification causes delays and increases fraud risks.

This workflow automates:

- KYC verification
- Fraud risk analysis
- Final SIM activation decision

Workflow Architecture




```
Customer Input
   ↓
KYC Verification Tool
   ↓
Gemini Fraud Detection Analysis
   ↓
Final Activation Decision
   ↓
SIM Activated / Manual Review / Rejection
```

State Management

TelecomState

The workflow state stores input data, intermediate processing outputs, and final decisions.

Input Fields

- customer_id
- aadhaar_number
- pan_number
- customer_location
- previous_sim_requests

Intermediate Outputs

- kyc_status
- fraud_risk
- fraud_reasoning

Final Outputs

- activation_decision
- final_message

Node 1: KYC Verification Tool

Validate Aadhaar and PAN details using rule-based checks.

Validation Logic

Aadhaar Validation

- Must contain exactly 12 digits

PAN Validation

- Must contain exactly 10 characters
- First 5 must be alphabets
- Next 4 must be digits
- Last character must be alphabet

Possible Outputs

- VERIFIED
- INVALID_DOCUMENT
- PENDING_VERIFICATION

Node 2: Fraud Detection using Gemini

Analyze fraud probability using:

- KYC verification result
- Customer location
- Previous SIM request history

Fraud Classification

- LOW_RISK
- MEDIUM_RISK
- HIGH_RISK

Node 3: Activation Decision Engine

Generate final activation decision based on fraud risk.

Decision Rules

Fraud Risk	Final Decision
LOW_RISK	SIM Activated

Fraud Risk	Final Decision
MEDIUM_RISK	Manual Review
HIGH_RISK	Reject Application

Testcases

```python
print("USE CASE 1: Telecom SIM Activation Workflow")
print("-" * 50)

customer_low_risk = {
    "customer_id": "CUST-001",
    "aadhaar_number": "123456789012",
    "pan_number": "ABCDE1234F",
    "customer_location": "Mumbai, Maharashtra",
    "previous_sim_requests": 1,
    "kyc_status": "",
    "fraud_risk": "",
    "fraud_reasoning": "",
    "activation_decision": "",
    "final_message": ""
}

print("Test Case 1: Valid Customer with Low SIM Request History")
result_1 = telecom_app.invoke(customer_low_risk)

print("\nFINAL REPORT - Customer 1")
print(f"KYC Status         : {result_1['kyc_status']}")
print(f"Fraud Risk         : {result_1['fraud_risk']}")
print(f"Fraud Reasoning    : {result_1['fraud_reasoning']}")
print(f"Activation Decision: {result_1['activation_decision']}")
print(f"Final Message      : {result_1['final_message']}")
```

```
USE CASE 1: Telecom SIM Activation Workflow
--------------------------------------------------
Test Case 1: Valid Customer with Low SIM Request History
[Node 1] Running KYC Verification...
KYC Status: VERIFIED
[Node 2] Running Gemini Fraud Detection Analysis...
Gemini Response:
RISK_LEVEL: LOW_RISK
REASONING: The customer's KYC status is verified, which significantly reduces the risk of identity fraud. Furthermore, having only one previous SIM request indicates normal customer behav
[Node 3] Making Final Activation Decision...
Decision: SIM Activated

FINAL REPORT - Customer 1
KYC Status         : VERIFIED
Fraud Risk         : LOW_RISK
Fraud Reasoning    : The customer's KYC status is verified, which significantly reduces the risk of identity fraud. Furthermore, having only one previous SIM request indicates normal cust
Activation Decision: SIM Activated
Final Message      : Customer CUST-001: SIM has been successfully activated.
```

2.

```python
customer_high_risk = {
    "customer_id": "CUST-002",
    "aadhaar_number": "INVALIDNUM",
    "pan_number": "WRONGPAN99",
    "customer_location": "Unknown Location",
    "previous_sim_requests": 8,
    "kyc_status": "",
    "fraud_risk": "",
    "fraud_reasoning": "",
    "activation_decision": "",
    "final_message": ""
}

print("Test Case 2: Invalid Documents with High SIM Request History")
result_2 = telecom_app.invoke(customer_high_risk)

print("\nFINAL REPORT - Customer 2")
print(f"KYC Status         : {result_2['kyc_status']}")
print(f"Fraud Risk         : {result_2['fraud_risk']}")
print(f"Fraud Reasoning    : {result_2['fraud_reasoning']}")
print(f"Activation Decision: {result_2['activation_decision']}")
print(f"Final Message      : {result_2['final_message']}")
```

```
Test Case 2: Invalid Documents with High SIM Request History
[Node 1] Running KYC Verification...
KYC Status: INVALID_DOCUMENT
[Node 2] Running Gemini Fraud Detection Analysis...
Gemini Response:
RISK_LEVEL: HIGH_RISK
REASONING: The "INVALID_DOCUMENT" KYC status indicates the customer's identity cannot be verified, which is a critical red flag. This is further compounded by an "Unknown Location" and a high numl
[Node 3] Making Final Activation Decision...
Decision: Reject Application

FINAL REPORT - Customer 2
KYC Status         : INVALID_DOCUMENT
Fraud Risk         : HIGH_RISK
Fraud Reasoning    : The "INVALID_DOCUMENT" KYC status indicates the customer's identity cannot be verified, which is a critical red flag. This is further compounded by an "Unknown Location" and a
Activation Decision: Reject Application
Final Message      : Customer CUST-002: Application rejected due to high fraud risk.
```

Use Case 2: Healthcare Appointment Prioritization Workflow

Problem Statement

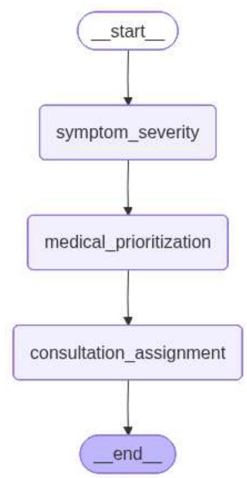
Hospitals must prioritize patients based on symptom severity and medical urgency. Manual triage may delay treatment for critical patients.

This workflow automates:

- Vital signs severity assessment
- Consultation priority classification
- Final doctor/department assignment

Workflow Architecture

```
Patient Input
   ↓
Symptom Severity Tool
   ↓
Gemini Medical Prioritization
   ↓
Consultation Assignment
   ↓
ICU / Specialist / General Physician
```



State Management

HealthcareState

Input Fields

- patient_id
- patient_name
- age
- symptoms
- fever_celsius
- oxygen_level
- heart_rate

- symptom_duration_hours
- existing_conditions

Intermediate Outputs

- severity_level
- priority_level
- priority_reasoning

Final Outputs

- consultation_type
- final_message

Node 1: Symptom Severity Tool

Classify severity based on vital signs.

Evaluation Parameters

- Body temperature
- Oxygen saturation (SpO2)
- Heart rate
- Symptom duration

Severity Levels

- STABLE
- MODERATE
- CRITICAL

Node 2: Medical Prioritization using Gemini

Evaluate consultation urgency using:

- Severity level
- Age
- Symptoms
- Existing medical conditions

Priority Levels

- EMERGENCY
- PRIORITY_CONSULTATION

- REGULAR_CONSULTATION

Node 3: Consultation Assignment

Objective

Assign the correct treatment path.

Decision Rules

Priority Level	Consultation Type
EMERGENCY	ICU / ER
PRIORITY_CONSULTATION	Specialist Doctor
REGULAR_CONSULTATION	General Physician

Test Cases

```python
print("USE CASE 2: Healthcare Appointment Prioritization Workflow")
print("-" * 50)

critical_patient = {
    "patient_id": "PAT-001",
    "patient_name": "John Doe",
    "age": 68,
    "symptoms": ["chest pain", "severe shortness of breath", "dizziness"],
    "fever_celsius": 39.8,
    "oxygen_level": 87,
    "heart_rate": 138,
    "symptom_duration_hours": 6,
    "existing_conditions": ["diabetes", "hypertension", "coronary artery disease"],
    "severity_level": "",
    "priority_level": "",
    "priority_reasoning": "",
    "consultation_type": "",
    "final_message": ""
}

print("Test Case 1: Elderly Patient with Critical Vitals and Multiple Conditions")
result_p1 = healthcare_app.invoke(critical_patient)

print("\nFINAL REPORT - Patient 1")
print(f"Severity Level     : {result_p1['severity_level']}")
print(f"Priority Level     : {result_p1['priority_level']}")
print(f"Clinical Reasoning : {result_p1['priority_reasoning']}")
print(f"Consultation Type  : {result_p1['consultation_type']}")
print(f"Final Message      : {result_p1['final_message']}")
```

```
USE CASE 2: Healthcare Appointment Prioritization Workflow
--------------------------------------------------
Test Case 1: Elderly Patient with Critical Vitals and Multiple Conditions
[Node 1] Running Symptom Severity Analysis...
Severity Level: CRITICAL
[Node 2] Running Gemini Medical Prioritization...
[Node 3] Assigning Final Consultation Type...
Consultation Type: ICU/ER

FINAL REPORT - Patient 1
Severity Level     : CRITICAL
Priority Level     : EMERGENCY
Clinical Reasoning : The patient presents with critical vital signs including severe hypoxia (SpO2 87%), significant tachycardia (138 BPM), and high fever
Consultation Type  : ICU/ER
Final Message      : Patient John Doe (ID: PAT-001): Immediate transfer to ICU/ER required.
```

2.

```
stable_patient = {
    "patient_id": "PAT-002",
    "patient_name": "Smith John",
    "age": 29,
    "symptoms": ["mild cough", "runny nose", "mild sore throat"],
    "fever_celsius": 37.8,
    "oxygen_level": 98,
    "heart_rate": 78,
    "symptom_duration_hours": 48,
    "existing_conditions": [],
    "severity_level": "",
    "priority_level": "",
    "priority_reasoning": "",
    "consultation_type": "",
    "final_message": ""
}

print("Test Case 2: Young Adult with Mild Upper Respiratory Symptoms")
result_p2 = healthcare_app.invoke(stable_patient)

print("\nFINAL REPORT - Patient 2")
print(f"Severity Level     : {result_p2['severity_level']}")
print(f"Priority Level     : {result_p2['priority_level']}")
print(f"Clinical Reasoning : {result_p2['priority_reasoning']}")
print(f"Consultation Type  : {result_p2['consultation_type']}")
print(f"Final Message      : {result_p2['final_message']}")
```

```
Test Case 2: Young Adult with Mild Upper Respiratory Symptoms
[Node 1] Running Symptom Severity Analysis...
Severity Level: STABLE
[Node 2] Running Gemini Medical Prioritization...
[Node 3] Assigning Final Consultation Type...
Consultation Type: General Physician

FINAL REPORT - Patient 2
Severity Level     : STABLE
Priority Level     : REGULAR_CONSULTATION
Clinical Reasoning : The patient presents with mild upper respiratory symptoms and stable vital signs, including normal oxygen saturation and heart rate, and a low-grade fever.
Consultation Type  : General Physician
Final Message      : Patient Smith John (ID: PAT-002): Scheduled for General Physician consultation.
```

3.

```
moderate_patient = {
    "patient_id": "PAT-003",
    "patient_name": "Sam Peter",
    "age": 52,
    "symptoms": ["persistent chest tightness", "fatigue", "mild breathlessness"],
    "fever_celsius": 37.2,
    "oxygen_level": 93,
    "heart_rate": 105,
    "symptom_duration_hours": 36,
    "existing_conditions": ["asthma"],
    "severity_level": "",
    "priority_level": "",
    "priority_reasoning": "",
    "consultation_type": "",
    "final_message": ""
}

print("Test Case 3: Middle-Aged Patient with Moderate Vitals and Pre-existing Asthma")
result_p3 = healthcare_app.invoke(moderate_patient)

print("\nFINAL REPORT - Patient 3")
print(f"Severity Level     : {result_p3['severity_level']}")
print(f"Priority Level     : {result_p3['priority_level']}")
print(f"Clinical Reasoning : {result_p3['priority_reasoning']}")
print(f"Consultation Type  : {result_p3['consultation_type']}")
print(f"Final Message      : {result_p3['final_message']}")
```

```
Test Case 3: Middle-Aged Patient with Moderate Vitals and Pre-existing Asthma
[Node 1] Running Symptom Severity Analysis...
Severity Level: MODERATE
[Node 2] Running Gemini Medical Prioritization...
[Node 3] Assigning Final Consultation Type...
Consultation Type: Specialist Doctor

FINAL REPORT - Patient 3
Severity Level     : MODERATE
Priority Level     : PRIORITY_CONSULTATION
Clinical Reasoning : The patient presents with persistent chest tightness and breathlessness, coupled with concerning vital signs including mild hypoxemia (SpO2 93%) and tachycardia (HR 105 BPM).
Consultation Type  : Specialist Doctor
Final Message      : Patient Sam Peter (ID: PAT-003): Priority appointment with Specialist Doctor within 2 hours.
```

Conclusion

The assignment successfully implements two production-style AI workflow systems using LangGraph and Gemini.

The Telecom workflow improves fraud prevention and onboarding efficiency by automating KYC and fraud analysis.

The Healthcare workflow improves patient triage by prioritizing consultations based on severity and medical urgency.

These implementations show how hybrid AI systems combining rule-based validation and LLM intelligence can solve complex operational challenges efficiently, accurately, and at scale.